

Corey's (*Python*) Spotify

WRAPPED

Listening Data Exploration by Corey Lopez

ABOUT

Student: Corey Lopez
Class: Comp 661

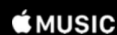
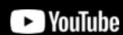
Dataset used: Last.fm scrobbles*

*Last.fm tracks the music you stream, creating listening reports, stats, and charts. View your personal artist, album, and charts from 7 days to all- time!

Export your listening data by requesting an API key, or create a CSV export!



CONNECT YOUR STREAMING SERVICE TO LAST.FM



PROJECT DESCRIPTION

OBJECTIVE: To scrape my listening data, clean and filter the dataset. and display Top Listening charts similar to Spotify's Wrapped

CONCEPTS: Importing *pandas*, load/reading **csv** files, convert/use *datetime* objects, receive/validate user input, and control loops!



DATASET

```
1 import pandas as pd
2 import datetime
3
4 try:
5     df = pd.read_csv("scrobbles.csv")
6 except FileNotFoundError:
7     print("Error: scrobbles.csv file not found.")
8
9 df.head()
```

✓ 0.1s Python

	uts	utc_time	artist	artist_mbid	album	album_mbid	track	track_mbid
0	1772768686	06 Mar 2026, 03:44	Ninajirachi	6897b46a-9236-4480-9ed7-300f6b85d62f	I Love My Computer	fc8bf81b-babb-47fe-99fc-841c1abfeb7c	Sing Good	95461b73-b0ab-4b33-9433-73b3ba3286d5
1	1772768488	06 Mar 2026, 03:41	Ninajirachi	6897b46a-9236-4480-9ed7-300f6b85d62f	I Love My Computer	fc8bf81b-babb-47fe-99fc-841c1abfeb7c	Battery Death	b2a9e5a9-278b-4818-be68-5b1be30d7f0e
2	1772768219	06 Mar 2026, 03:36	Ninajirachi	6897b46a-9236-4480-9ed7-300f6b85d62f	I Love My Computer	fc8bf81b-babb-47fe-99fc-841c1abfeb7c	Infohazard	70c821ed-05f7-4151-975f-40e209a9c6a7
3	1772768036	06 Mar 2026, 03:33	Ninajirachi	6897b46a-9236-4480-9ed7-300f6b85d62f	I Love My Computer	fc8bf81b-babb-47fe-99fc-841c1abfeb7c	All I Am	bce433ec-9392-474a-95e5-078d4bb191e3
4	1772767969	06 Mar 2026, 03:32	Ninajirachi	6897b46a-9236-4480-9ed7-300f6b85d62f	I Love My Computer	fc8bf81b-babb-47fe-99fc-841c1abfeb7c	ⓂⓂⓂⓂⓂⓂ	f0807482-0c84-4dd9-acf5-4e5e317b7194

This dataset contains my personal music listening history collected from Last.fm, where each row counts one scrobble.

Surprisingly, I've tracked my scrobbles since 2018!

```
3 df.info()
✓ 0.0s

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 92700 entries, 0 to 92699
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype
---  -
0   uts         92700 non-null  int64
1   utc_time    92700 non-null  object
2   artist      92700 non-null  object
3   artist_mbid 76376 non-null  object
4   album       92637 non-null  object
5   album_mbid  64400 non-null  object
6   track       92695 non-null  object
7   track_mbid  65679 non-null  object
dtypes: int64(1), object(7)
memory usage: 5.7+ MB
```

df.info() tells me I have 92,700 scrobbles!
There are 7 columns with int and object types.

In order to perform analysis, I want to convert the **'uts'** column into datetime.
I initially imported datetime, but I can do this with *pandas*!

```
1 df['datetime'] = pd.to_datetime(df['uts'], unit='s')
2 df.dtypes
3 df.head(1)
✓ 0.0s
```

	uts	utc_time	artist	artist_mbid	album	album_mbid	track	track_mbid	datetime
0	1772768686	06 Mar 2026, 03:44	Ninajirachi	6897b46a-9236-4480-9ed7-300f6b85d62f	I Love My Computer	fc8bf81b-babb-47fe-99fc-841c1abfeb7c	Sing Good	95461b73-b0ab-4b33-9433-73b3ba3286d5	2026-03-06 03:44:46

Python

Next, I want to:

- 1) drop unnecessary columns using df.drop()
- 2) extract day, month and year from these datetime objects and make those new columns
- 3) move the datetime and other columns from the end to the beginning.

```
1 df = df.drop(columns=['uts', 'utc_time', 'artist_mbid', 'album_mbid', 'track_mbid'])
2
3 df['year'] = df['datetime'].dt.year
4 df['month'] = df['datetime'].dt.month
5 df['day'] = df['datetime'].dt.day
6
7 df = df[
8     [
9         'datetime',
10        'year',
11        'month',
12        'day',
13        'track',
14        'artist',
15        'album' ]
16 ]
17 df.head(1)
```

✓ 0.0s

	datetime	year	month	day	track	artist	album
0	2026-03-06 03:44:46	2026	3	6	Sing Good	Ninajirachi	I Love My Computer

perfect!

Quality check! I know I switched to Apple Music at some point. Do I have full years of valid data?

```
1 df_sorted = df.sort_values('datetime')
2
3 first_streams = df_sorted.groupby('year').first()
4
5 first_streams['datetime']
```

✓ 0.0s

```
year
2018    2018-08-23 01:13:51
2022    2022-12-01 19:39:52
2023    2023-01-01 00:50:44
2024    2024-01-01 00:00:19
2025    2025-01-01 00:01:02
2026    2026-01-01 04:19:12
Name: datetime, dtype: datetime64[ns]
```

I can see I took a break from scrobbling consistently until 2023.
Let's filter for 2023-2026 only!

```
df_analysis = df[(df['year'] >= 2023) & (df['year'] <= 2026)]
```

FUNCTIONS

```
def filter_year(year):  
    """  
    Return a filtered data frame containing only scrobbles from the given year.  
    """  
    data = df_analysis[df_analysis['year'] == year]  
  
    if data.empty:  
        print("No data found for that year.")  
  
    return data
```


FUNCTIONS

```
def top_artists(year):  
    """  
    Display top 3 most played ARTISTS for a given year.  
    """  
    data = filter_year(year)  
  
    #count artists and return the top 3  
    results = (data['artist'].value_counts().head(3))  
  
    print("\nTop 3 Artists in", year)  
    print(results)
```

```
def top_albums(year):  
    """  
    Display top 3 most played ALBUMS for a given year.  
    """  
    data = filter_year(year)  
  
    #count values and return the top 3  
    results = (data['album'].value_counts().head(3))  
  
    print("\nTop 3 Albums in", year)  
    print(results)
```

FUNCTIONS

```
def top_songs(year):  
    """  
    Display the top 3 most played SONGS for a given year.  
    """  
    data = filter_year(year)  
  
    #count TRACKS and return the top 3  
    results = (data['track'].value_counts().head(3))  
  
    print("\nTop 3 Songs in", year)  
    print(results)  
  
#an idea for a future iteration, how can I ensure I'm not counting different songs by the same name?
```

USER INPUT MENU

```
1 print("Welcome to Corey's Spotify Wrapped!")
2
3 while True:
4
5     print('\n Which category would you like to see?')
6     print("1 - Top 3 Artists")
7     print("2 - Top 3 Albums")
8     print("3 - Top 3 Songs")
9     print("4 - Exit")
10
11     choice = input("Choose an option: ")
12
13     if choice == "4":
14         print("Happy Spotify Wrapped!")
15         break
16
17     try:
18         year = int(input("Enter a year (2023-2026): "))
19
20         if year < 2023 or year > 2026:
21             raise ValueError("Year out of range")
22
23     except ValueError:
24         print("Please enter a valid year between 2023 and 2026.")
25
26         continue
27
28     if choice == "1":
29         top_artists(year)
30
31     elif choice == "2":
32         top_albums(year)
33
34     elif choice == "3":
35         top_songs(year)
36
37     else:
38         print("Invalid option")
```

USER INPUT MENU

Sample console output:

```
Welcome to Corey's Spotify Wrapped!

Which category would you like to see?
1 - Top 3 Artists
2 - Top 3 Albums
3 - Top 3 Songs
4 - Exit

Top 3 Albums in 2024
album
eternal sunshine      908
COWBOY CARTER         545
BRAT                   505
Name: count, dtype: int64

Which category would you like to see?
1 - Top 3 Artists
2 - Top 3 Albums
3 - Top 3 Songs
4 - Exit
Thanks for using Corey's Python Wrapped!
Bye!
```

**THANKS FOR USING
COREY'S
PYTHON-POWERED
SPOTIFY WRAPPED !!!**

